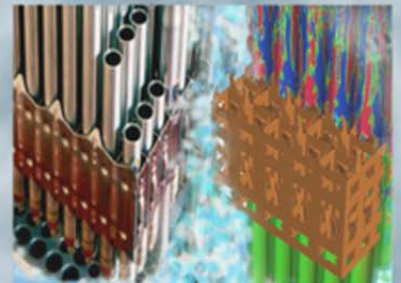
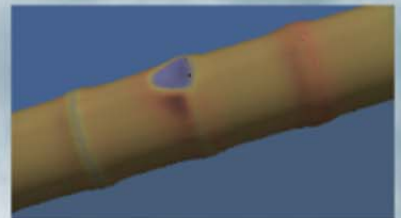
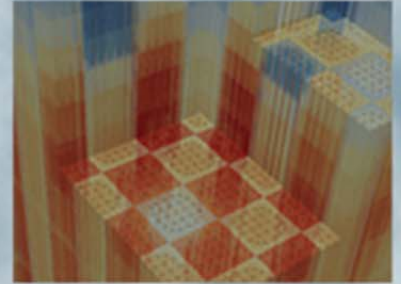


Integrating BISON into VERA-CS and Fuel Temperature Calculations

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EXECUTIVE SUMMARY

Approaches to integrating BISON into VERA-CS were identified as: 1) Coupled TIAMAT calculations in which BISON is solved simultaneously with MPACT and CTF via direct coupling, 2) Inline TIAMAT in which BISON calculations are performed as part of the same VERA-CS calculation sequence as coupled MPACT-CTF calculations but there is no feedback from the BISON calculations into the MPACT-CTF calculations, and 3) Stand-alone (“Offline”) BISON calculations in which desired data is parsed from the results of an already-completed MPACT-CTF VERA-CS analysis and BISON calculations are performed afterwards. The current maximum capabilities of these approaches in the latest VERA build are a single ramp-to-power for Coupled TIAMAT, analysis of two subsequent operating cycles for a single pin in Inline TIAMAT, and BISON analysis for every fuel pin in a PWR quarter-core for multiple operating cycles in Stand-alone BISON.

The goal of fully-coupled VERA-CS calculations would be to get better physics into BISON for better fuel calculations (e.g., detailed power histories and fast flux from MPACT) and use better fuel temperatures in MPACT for depletion. While Inline TIAMAT and stand-alone BISON calculations may prove to be good enough for simple core-follow depletion analyses, specific use cases (e.g., detailed aspects of CASL challenge problems or uncertainty quantification investigations into inaccuracies caused by uncoupled calculations) would require or benefit from Coupled TIAMAT.

Work planning efforts during late FY16 or early FY17 will assess what coupling physics is actually needed in Coupled TIAMAT (i.e., what data needs to be passed back and forth between MPACT and BISON and which parameters should convergence be enforced for) and how that coupling physics fits into the MPACT predictor-corrector scheme. Future work may also include quantifying the accuracy improvements and runtime increases associated with Coupled TIAMAT compared to Inline TIAMAT for stand-alone BISON. Further efforts investigate reducing the computational burden of BISON calculations in Coupled TIAMAT by changing modeling approximations or assumptions.

Improved fuel temperature tables have been generated using BISON with significantly more data points over time compared to previous work and an increased peak burnup of 49 GWd/MTU. These new fuel temperatures appear to offer both qualitative and quantitative improvements over previous BISON fuel temperatures, and qualitatively agree with FRAPCON calculations for the same conditions. The use of these new BISON fuel temperature tables will be demonstrated in VERA-CS depletion analyses to reevaluate Watts Bar Nuclear Plant Unit 1 Cycles 1–12 in the near future. The goal of using improved BISON fuel temperatures is to increase accuracy in parameters such as soluble boron concentration and local power distributions.

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1. INTRODUCTION

The primary objective of this milestone is to work with PHI to assist with efforts coupling BISON into VERA-CS via direct coupling (e.g., Tiamat) or stand-alone calculations performed parsing data from existing output files to create new ones (e.g., running standalone BISON calculations constructed by parsing power histories from VERA-CS calculations). It is expected that FY16 efforts in this area will identify promising paths forward and demonstrate some success in applying one or more approach (e.g., running batch or Multiapp-based BISON jobs), but it is not expected that FY16 efforts in this area will result in BISON being fully coupled into VERA-CS; such efforts are considered to be future work beyond the scope of this year.

A secondary objective of this milestone is to work with PHI to identify pathways to using BISON to calculate fuel temperatures for use in VERA-CS depletion calculations. If possible, one or more pathway should be demonstrated and implemented fully for fuel temperature calculations, allowing VERA-CS users to generate BISON-calculated fuel temperatures as a function of operational parameters such as linear heat rate (LHR) and specific burnup for use in neutronics (MPACT or MPACT-CTF) core-follow depletion calculations.

The work performed for this milestone followed a five-step execution plan:

1. Work with PHI to identify one or more promising pathways to coupling or integrating BISON (2D R-Z) into VERA-CS.
2. Work with PHI to demonstrate proof-of-principle implementation of one or more pathway identified.
3. Work with PHI to identify one or more promising pathways to using BISON to calculate fuel temperatures as a function of linear heat rate and burnup for neutronics depletion calculations in VERA-CS.
4. Work with PHI to demonstrate one or more pathway to BISON-calculated fuel temperatures, implementing it in a user-friendly manner if possible.
5. Work with PHI to plan future work that will demonstrate fully coupled BISON calculations in VERA-CS.

Completion criteria were established for each step of the execution plan. This milestone report divides the execution steps into those associated with integrating BISON into VERA-CS and those associated with generating fuel temperatures for VERA-CS core follow depletion calculations. This report provides information and results which satisfy all of the completion criteria established for the execution steps: pathways identified for integrating 2D R-Z BISON calculations into VERA-CS are documented with results-to-date show for several approaches, pathways identified for using BISON to calculate fuel temperatures as a function of LHR and burnup for core-follow calculations are described and relevant results are shown, and finally future work is identified for the remainder of FY16 and into FY17 to expand the integration of BISON into VERA-CS including expanding use and usefulness of fully-coupled calculations with MPACT-CTF and BISON.

2. INTEGRATING BISON INTO VERA-CS

Past analysis efforts have focused on coupled MPACT and Cobra-TF (CTF) calculations as the major codes utilized in VERA-CS. However, Fiscal year 2015 (FY15) efforts such as depletion analysis performed for Watts Bar Nuclear Plant Unit 1 (WBN1) Cycles 1–12 [1] began to incorporate the use of BISON as the progressing maturity of the code improved its use and the benefits including BISON in VERA-CS analyses (e.g., improved fuel temperature predictions) became more important to the problems being analyzed. Past work in CASL already started integrating BISON into VERA-CS via TIAMAT.

Three approaches to integrating (or coupling) BISON into VERA-CS have been identified as worth of pursuing: 1) Coupled TIAMAT calculations in which BISON is solved simultaneously with MPACT and CTF via direct coupling, 2) Inline TIAMAT in which BISON calculations are performed as part of the same VERA-CS calculation sequence as coupled MPACT-CTF calculations but there is no feedback from the BISON calculations into the MPACT-CTF calculations, and 3) Stand-alone (“Offline”) BISON calculations in which desired data is parsed from the results of an already-completed MPACT-CTF VERA-CS analysis and BISON calculations are performed afterwards. Table 1 summarizes some of the known benefits and challenges of each approach. All three approaches support the use of simplified VERAIn-based input creation and the use of VERAView for visualizing results.

Table 1. Benefits and challenges for Coupled TIAMAT, Inline TIAMAT, and stand-alone BISON

Approach	Benefits	Challenges
Coupled TIAMAT (MPACT-CTF-BISON)	Highest accuracy physics modeling and results	Computational cost; Syncing time grid, spatial mesh, and data; Which parameters do you couple and converge on?; BISON restarts vs. full calculations each step?
Inline TIAMAT	More efficient than stand-alone BISON calculations	No feedback during calculations; BISON restarts vs. full calculations each step?
“Offline”/Stand-alone BISON calculations through VERAIn	Lowest computation cost; Simplest approach; Eliminates coupling issues	No feedback during calculations

Coupled TIAMAT calculations were demonstrated for a “minicore” problem during FY15 efforts in CASL. Ongoing efforts in TIAMAT are updating the BISON template, models, and solvers used during the coupled calculations. Current use of Coupled TIAMAT is restricted to an initial ramp to power; depletion does not current work. A simple single-pin analysis using the latest build of VERA has been performed to demonstrate some of the current capabilities of Coupled TIAMAT. Results are shown in Figure 1 for total rod power, average fuel temperature, and average pellet-cladding gap thickness.

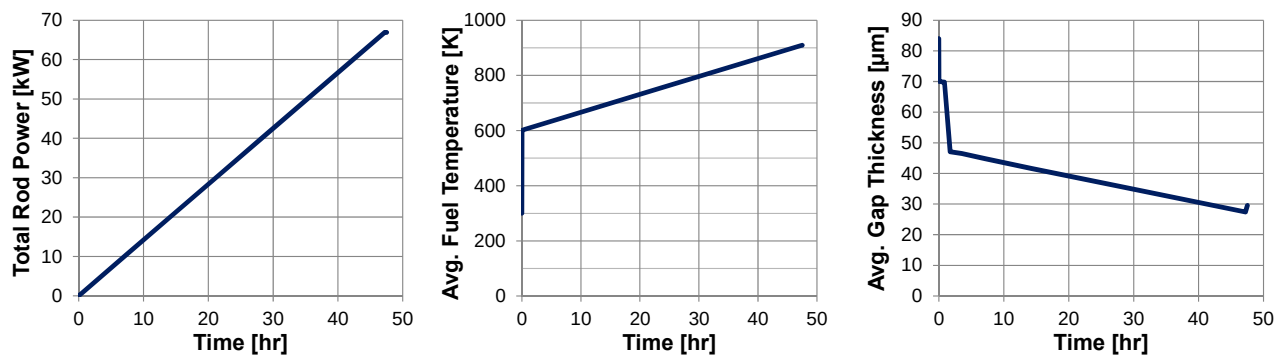


Figure 1. Results for total rod power (left), average fuel temperature (center), and average gap thickness (right) for a single-pin ramp-to-power calculation performed in Coupled TIAMAT

Inline TIAMAT runs MPACT/CTF for one time step and then, after completing that calculation, simultaneously starts a BISON calculation for that time step and an MPACT/CTF calculation for the subsequent time step. Inline TIAMAT can currently run a ramp-to-power and steady operation for Cycle 1, down-power followed by zero-power hold for a refueling outage, and then an up-ramp and steady operation for Cycle 2. Inline TIAMAT should allow some improvements and enable developers and users to start seeing the impacts of integrating BISON into coupled calculations. A single-pin, single-cycle Inline TIAMAT analysis was performed using the latest build of VERA to demonstrate its current capabilities. Results from this Inline TIAMAT analysis are shown in Figure 2 for total rod power, average fuel temperature, and average gap thickness.

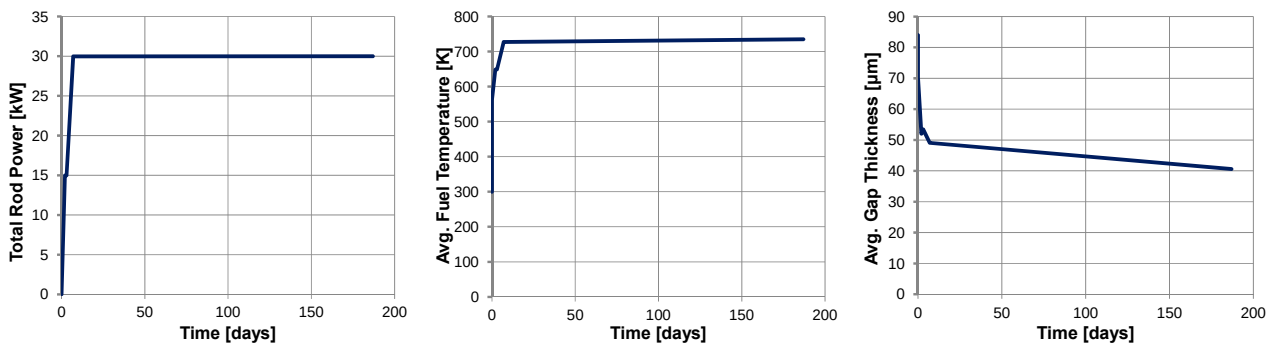


Figure 2. Results for total rod power (left), average fuel temperature (center), and average gap thickness (right) for a single-pin single-cycle analysis performed using Inline TIAMAT

The capability and use of stand-alone (“offline”) BISON calculations was demonstrated at length for a recent CASL milestone report in which BISON stand-alone calculations were performed for each pin in a quarter-core of WBN1 Cycles 1–3 using power histories and boundary conditions parsed from the MPACT-CTF depletion results. [2] Further details may be found in the original document. Demonstration runs at scale for PCI assessment will also be completed during FY16 using stand-alone BISON.

A Level 3 milestone within PHI being completed in parallel to this milestone will further describe both Coupled TIAMAT and Inline TIAMAT and should demonstrate further results. Once Inline TIAMAT and stand-alone BISON calculations are both working, significant efforts will be spent assessing what coupling physics is actually needed in future versions of Coupled TIAMAT (i.e., what data needs to be passed back and forth between MPACT and BISON), which parameters convergence should be enforced upon, and how the coupling physics fits into the MPACT predictor-corrector scheme. This work planning effort will be finished late in FY16 or early FY17. The goal of

fully-coupled VERA-CS calculation would be to get better physics into BISON for better fuel calculations (e.g., detailed power histories and fast flux from MPACT) and use better fuel temperatures in MPACT for depletion. While this fully-coupled solve may prove unnecessary for relatively simple core-follow type depletion analyses that would be of most direct use to industry, specific use cases (e.g., detailed aspects of CASL challenge problems, uncertainty quantification investigations into inaccuracies caused by uncoupled calculations) would require or benefit from Coupled TIAMAT.

3. GENERATING FUEL TEMPERATURES WITH BISON

Previous VERA-CS depletion analyses used fuel temperature tables generated based on results from nine single-pin BISON calculations, based on three rod-average LHRs evaluated using three axial power distributions each. [1] The temperature difference (ΔT) between the average fuel temperature and bulk coolant temperature at a given axial node was calculated to assess ΔT as a function of LHR and burnup. Challenges encountered limited the previous results to only a few points in time during depletion and a maximum burnup value of 22.5 GWd/MTU.

Subsequent BISON calculations generated improved fuel temperature tables with significantly more data points over time and a peak burnup of 49 GWd/MTU. These BISON calculations varied the rod-average LHR for a part-length pin while assuming a uniform axial power distribution across the pin and a fixed clad surface temperature. Using an updated version of BISON and the SuperLU solver from PETSc version 3.5.3, these calculations achieved much better convergence results, though problems still occurred when running high LHRs at high burnup. In addition to these new BISON results, fuel temperatures were also generated using the FRAPCON fuel performance code and CTF for comparative purposes. The CTF calculations assumed a single gap conductance value throughout burnup. Overall, Figure 3 shows that the new BISON fuel temperatures appear to offer qualitative improvements compared to the old BISON temperatures due to changes in modeling approximations and parameter values. The new BISON results agree qualitatively with those from FRAPCON as well, though the beginning of life (BOL) fuel temperatures quantitatively differ somewhat. Results from CTF calculations with two different gap conductance values both appear higher than the BISON results but offer some qualitative promise. Ongoing CTF development to enable the use of a burnup-dependent gap conductance and enhanced physics models (e.g., fuel relocation and thermal expansion) may further improve those predictions.

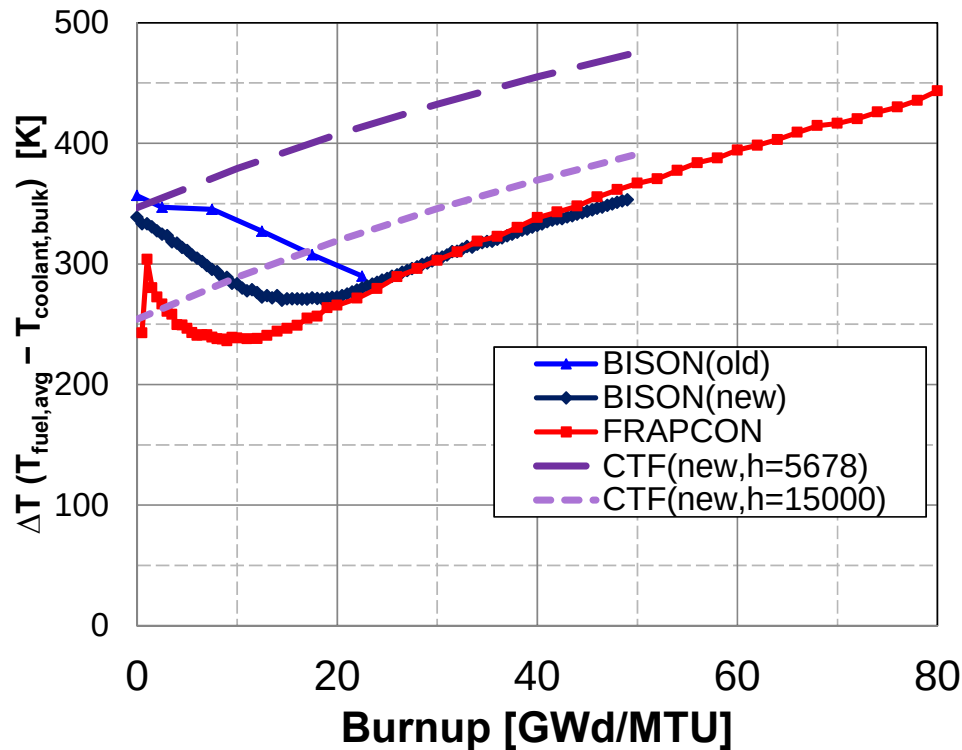


Figure 3. ΔT as a function of burnup (in units of GWd/MTU) for an LHR of 5.64 kW/ft (185 W/cm).

A preliminary evaluation of IFBA vs non-IFBA fuel rods using both BISON and FRAPCON indicates that relatively small but noticeable differences in fuel temperatures are observable; therefore, full-scale use of IFBA models in BISON will likely be needed in the near future. Recent BISON development efforts deploying IFBA models for UO₂ fuel should adequately support the initial use of this, though subsequent work may be needed for items such as using the internal mesh generation script in BISON to generate a mesh containing both solid and annular fuel pellets in the same stack.

The use of these new BISON fuel temperature tables will be demonstrated in VERA-CS depletion analyses to reevaluate Watts Bar Nuclear Plant Unit 1 Cycles 1–12 in the near future; these calculations are currently in-progress.

4. CONCLUSIONS

Approaches to integrating BISON into VERA-CS were identified as: 1) Coupled TIAMAT calculations in which BISON is solved simultaneously with MPACT and CTF via direct coupling, 2) Inline TIAMAT in which BISON calculations are performed as part of the same VERA-CS calculation sequence as coupled MPACT-CTF calculations but there is no feedback from the BISON calculations into the MPACT-CTF calculations, and 3) Stand-alone (“Offline”) BISON calculations in which desired data is parsed from the results of an already-completed MPACT-CTF VERA-CS analysis and BISON calculations are performed afterwards. The current maximum capabilities of these approaches in the latest VERA build are a single ramp-to-power for Coupled TIAMAT, analysis of two subsequent operating cycles for a single pin in Inline TIAMAT, and BISON analysis for every fuel pin in a PWR quarter-core for multiple operating cycles in Stand-alone BISON.

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